Question I

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix and let $q_1 \in \mathbb{R}^n$ satisfy $\|q_1\|_2 = 1$. Consider the following iteration:

$$
\begin{aligned}
&r_0 = q_1, \quad \beta_0 = 1, \quad q_0 = 0, \quad k := 0\\
&\textbf{while } \ \beta_k \neq 0\\
&\quad q_{k+1} := r_k/\beta_k\\
&\quad k := k + 1\\
&\quad \alpha_k := q_k^T A q_k\\
&\quad r_k := (A - \alpha_k I)q_k - \beta_{k-1} q_{k-1}\\
&\quad \beta_k := \|r_k\|_2\\
&\textbf{end}
\end{aligned}
$$

Let $K_k = \text{span}\{q_1, Aq_1, \cdots, A^{k-1}q_1\}$ and $Q_k = [q_1 \cdots q_k]$ be a matrix whose columns are $\{q_i\}_{i=1}^k$ obtained from the above iteration.

1. Assume that the iteration does not terminate. Show that Q_k has orthonormal columns, and that they span K_k.

2. What is the purpose of this algorithm? Justify your answer.

Question II

Consider the oscillatory second order ordinary differential equation (ODE)

$$y''(t) + \lambda^2\, y(t) + g(y(t)) = 0, \qquad 0 < t \leq T,$$

with the initial data

$$y(0) = \alpha, \qquad y'(0) = \beta,$$

where $\lambda >> 1$, α and β are given constants, and $g(y)$ is a given Lipschitz continuous function.

Choose a time step $\tau > 0$ and denote $t_n = n\tau$ for $n \geq 0$. Let y^n be the numerical approximation of $y(t_n)$ for $n \geq 0$.

1. Re-write the above ODE into its equivalent integral formulation near $t = t_n$ with $t = t_n + s$ for $s \in \mathbb{R}$ via the variation-of-constant formula.

2. Based on the integral formulation, design the following time integrator via proper numerical quadratures

$$y^{n+1} = 2\cos(\lambda\tau)y^n - y^{n-1} - \frac{\sin(\lambda\tau)}{\lambda} g(y^n), \qquad n \geq 1,$$

with

$$y^0 = \alpha, \qquad y^1 = \alpha\cos(\lambda\tau) + \frac{\beta}{\lambda}\sin(\lambda\tau) - \frac{\sin(\lambda\tau)}{2\lambda} g(\alpha).$$

Under proper stability assumption, prove the following error bound

$$|y(t_n) - y^n| \leq C\tau^2, \qquad 0 \leq n \leq \frac{T}{\tau},$$

where $C > 0$ is a constant independent of τ.

Question III

Consider an energy functional for $\rho \in \mathcal{P}(\mathbb{R})$ (probability distribution on the real line), given by

$$F[\rho] = \int_{\mathbb{R}\times\mathbb{R}} \rho(x)K(x-y)\rho(y)\,\mathrm{d}x\,\mathrm{d}y + \int_{\mathbb{R}} \rho(x)(1-\rho(x))\,\mathrm{d}x + \int_{\mathbb{R}} \rho(x)\ln\rho(x)\,\mathrm{d}x,$$

where $K : \mathbb{R} \to \mathbb{R}_+$ is a given kernel.

1. Write down explicitly the Euler-Lagrange equation corresponding to F.
2. Consider the dynamics

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho(x,t) = \nabla \cdot \Big(\rho(x,t)\nabla\frac{\delta F}{\delta \rho}\big[\rho(x,t)\big]\Big).$$

Show that $F[\rho(x,t)]$ is decreasing in t.